



Team Details

Team Name:

Web Shooters

SR. NO	ROLE	NAME	ACADEMIC YEAR
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Problem Statement Addressed



KLA PS01 — AI-Based Restoration of Degraded Semiconductor Inspection Images

DESCRIPTION / DETAILS

- Inspection images decide whether a wafer passes: one pixel of noise can hide a killer defect.
- Three degradations, order undisclosed — speckle noise, additive Gaussian noise, 2× downsampling.
- Input 128×128 noisy → output 256×256 clean (also 256→512). Grayscale float32.
- Degraded pixels fall outside [0,1] by design; ground truth always inside it.
- Scored on three axes: PSNR + SSIM + LPIPS, H100 end-to-end throughput, training/compute hygiene.
- Hidden test set contains out-of-distribution content — must generalise, not memorise.



Idea Description – Describe your Idea/Solution/Prototype



One small fully-convolutional network denoises and 2× super-resolves in a single pass, built on a degradation model recovered from the data rather than guessed.

KEY CONCEPT & APPROACH

- Reverse-engineered KLA's exact degradation recipe from the 3200 provided pairs.
- That yields unlimited correctly-degraded training data, with fresh noise every epoch.

SOLUTION OVERVIEW

- 1.37M-parameter residual CNN; all work at low resolution, one PixelShuffle at the end — 4× cheaper.
- Fully convolutional, so the same weights restore 128→256 and 256→512.



Proposed Solution – Describe your Idea/Solution/Prototype



Measured forward model → unlimited synthetic pairs → compact CNN → loss blended for pixel and perceptual quality.

SOLUTION DETAILS

- DEGRADATION RECOVERED (measured): 2×2 area-average downsample; noise applied after it; speckle multiplicative, $\text{var} \propto \text{pixel}^2$ ($r = 0.993$); $\sigma_{\text{speckle}} 0.10\text{--}0.25$, $\sigma_{\text{gauss}} 0.00\text{--}0.15$.
- `verify_degradation.py` re-derives every constant from the data and ABORTS on mismatch.
- TRAINING: 50/50 real and synthesised pairs; noise ranges widened beyond measured; 8-fold dihedral augmentation.
- MODEL: 16 residual blocks $\times$ 64 ch $\rightarrow$ PixelShuffle $\times 2$. No global skip from the noisy input. Output clamped to $[0,1]$.
- LOSS: Charbonnier blended with an LPIPS perceptual term by weight interpolation of two checkpoints.
- STACK: PyTorch 2.9.1+cu128, trained on NVIDIA GeForce GTX 1650 (4 GiB) — base 110 ep/5.4 h + perceptual 25 ep/1.4 h.
- FEASIBILITY: 16 MB checkpoint, 31.5 ms/image end-to-end; runs unchanged on CPU or any NVIDIA GPU.



Innovation and Uniqueness



We measured what others assume — which produced both the data advantage and the model.

KEY INNOVATION

- Degradation recovered, not guessed — giving unlimited perfectly-matched training data.
- Noise SPECTRUM matched, not just variance: caught a 16% high-frequency error every variance test passed.
- Perceptual/pixel weight interpolation — 93% of the LPIPS gain for 2.9% of the PSNR gain, zero extra training.

COMPETITIVE ADVANTAGE

- Beats bicubic on 200/200 validation images — every one, not on average.
- Deliberately small (1.37M params): KLA scores throughput, and oversized models lose it.
- Ideas tested and REJECTED on evidence: 8× TTA (worsened LPIPS), synthetic-pattern training (-0.22 dB).



Impact and Benefits



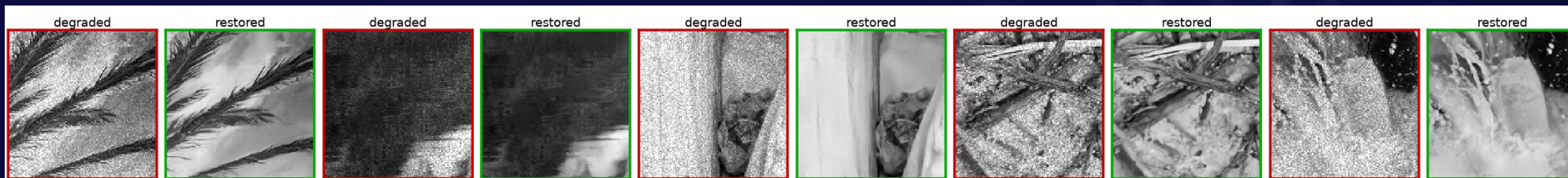
Recovers detail engineers currently accept as lost — and generalises to structure it never trained on.

Primary Impact

- PSNR 23.23 → 28.38 dB
- SSIM 0.5395 → 0.7477
- LPIPS 0.4369 → 0.2158
- Untrained control 11.70 dB — the metric can fail.

Quantifiable Outcomes

- +5.14 dB PSNR, +0.2083 SSIM, 0.2211 better LPIPS
- 31.5 ms/image end-to-end (read → preprocess → transfer → model → save)
- Unseen semiconductor-like structure: +2.56 dB mean, ahead on 14/15 patterns





GitHub & Video Link



GitHub Repository

https://github.com/bevinj2255/i4c_hackathon



Prototype / Simulation Video

Not submitted — full results, figures and code are in the repository.

REFERENCES

Lim et al., EDSR, CVPRW 2017 · Shi et al., PixelShuffle, CVPR 2016 · Zhang et al., LPIPS, CVPR 2018 · Zhai et al., IEEE Access 11:21049, 2023 · Terven et al., Artif Intell Rev 58:195, 2025 · Monga et al., IEEE SPM 38(2), 2021.

No external datasets or pretrained weights were used for the model.